DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION- 2025-2026 , PERIODIC ASSESSMENT – 2

SET–I

SUBJECT - MATHEMATICS

Time: 3 hrs. M. M: 80

Date : _____ Name: _____ Class X / Sec _____

General Instructions:

- General Instructions: Read the following instructions carefully and follow them
- 1. This question paper contains 38 questions. All Questions are compulsory.
- 2. This Question Paper is divided into 5 Sections A, B, C, D and E.
- 3. In Section A, Question numbers 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
- 4. In Section B, Question numbers 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
- 5. In Section C, Question numbers 26-31 are short answer (SA) type questions, carrying 03 marks each.
- 6. In Section D, Question numbers 32-35 are long answer (LA) type questions, carrying 05 marks each.
- 7. In Section E, Question numbers 36-38 are case study-based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
- 8. There is no overall choice. However, an internal choice in 2 questions of Section B, 2 questions of Section C and 2 questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
- 9. Draw neat and clean figures wherever required. Take $\pi = 22/7$.
- 10. Use of calculators is not allowed.

SECTION A

Directions: Question no. 1 to 18 are MCQs and question 19 and 20 are assertion and reason type each consist of 1 mark. (1*18=18)

MULTIPLE CHOICE BASED QUESTIONS

1. If $a = 2^2 \times 3x$, $b = 2^2 \times 3 \times 5$, $c = 2^2 \times 3 \times 7$ and LCM $(a,b,c) = 3780$, then x is equal to ___

(A) 1 (B) 2 (C) 3 (D) None of these

2. If α and β are the zeroes of polynomial $f(x) = px^2 - 2x + 3p$ and $\alpha + \beta = \alpha\beta$, then p is___

(A) $-\frac{2}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

3. If graph of a polynomial does not intersect the x-axis but intersects y-axis in one point, then no. of real zeroes of the polynomial is equal to_____

(A) 0 (B) 1 (C) 0 or 1 (D) none of these

$\frac{-b}{a} = \frac{c}{a} = \frac{2}{p} = \frac{3\cancel{p}}{\cancel{p}}$

$= p = 2/3$

4. . If the pair of equations $ax+by=c$ and $dx+ey=f$ are inconsistent, which of the following must be true?

A) $\frac{a}{d} \neq \frac{b}{e}$ B) $\frac{a}{d} = \frac{b}{e} = \frac{c}{f}$ C) $\frac{a}{d} = \frac{b}{e} \neq \frac{c}{f}$ D) $\frac{a}{d} \neq \frac{b}{e} = \frac{c}{f}$

5. Which method is NOT used to solve a pair of linear equations?

A) Graphical method
B) Substitution method
C) Elimination method
D) Factorization method

6. A teacher asks four students to answer necessary condition(s) of quadratic equations: For $ax^2 + bx + c = 0$, Who answered wrong?

(i) Ria answers If $b^2 - 4ac$ is a perfect square, the roots are rational.
(ii) Aryan answers If $b^2 = 4ac$, the roots are real and equal.
(iii) Ryan Answers If $b^2 - 4ac$ is negative, no real roots exist.
(iv) Mala answers If $b^2 = 4ac$, the roots are real and unequal

$b^2 - 4ac \geq 0$

(A) RIA (B) ARYAN (C) RYAN (D) MALA

7. Positive value of p for which equation $x^2 + px + 64 = 0$ and $x^2 - 8x + p = 0$ will both have real roots will be ____
(A) $p \geq 16$ (B) $p \leq 16$ (C) $p = 16$ (D) none of these

8. If zeroes of the quadratic polynomial $ax^2 + bx + c$ $(a, c \neq 0)$ are equal, then___
(A) c and b must have opposite signs
(B) c and a must have opposite signs
(C) c and b must have same signs
(D) c and a must have same signs

9. If (2x+2), 6x and (2x+1) are in A.P. Find the value of 2x.
(A)3/8
(B) 3/4
(C)3/6
(D) 6/10

10. A card is drawn from a deck of 52 cards. The event E is that card is not an ace of hearts. The number of outcomes favourable to E is

(A) 4 (B) 13 (C) 48 (D) 51

11. If 7th and 13th term of an A.P. are 34 and 64 respectively, then find the difference between first term and common difference_____
(A) 1 (B) 8 (C) 9 (D) 3

12. If the centre of a circle is (3, 5) and end points of a diameter are (4, 7) and (2, y), then the value of y is_____.
(A)3 (B) -3 (C) 7 (D) 4

13. The point P on x -axis equidistant from the points A (- 1, 0) and B (5, 0) is_____.
(A)(2, 0) (B) (0, 2) (C) (3, 0) (D) (- 3, 5)

14. The 4th term from the end of an AP -11, -8, -5, ..., 49 is_____.
(A)37 (B) 40 (C) 43 (D) 58

$x^2 + 16x + 64$
$x^2 + 8x + 8x + 64$

15. If $\alpha+\beta=90^\circ$ and $\alpha=2\beta$, then $\cos^2\alpha+\sin^2\beta$ is equal to

(a) 1 (b) $\frac{1}{2}$ (c) 0 (d) 2

16. In triangle ABC, $\angle BAC = 90^\circ$ and $AD \perp BC$. Then

(A) $BD.CD = BC^2$ (B) $AB.AC = BC^2$

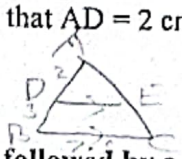
(C) $BD.CD = AD^2$ (D) $AB.AC = AD^2$

17. If ΔABC is right angled at C. then the value of $\sec(A+B)$ is:

(A) 0 (B) 1 (C) $\frac{2}{\sqrt{3}}$ (D) Not defined

18. D and E are respectively the points on the sides AB and AC of a triangle ABC such that AD = 2 cm, BD = 3 cm, BC = 7.5 cm and DE || BC. Then, length of DE (in cm) is

(A) 2.5 (B) 3 (C) 5 (D) 6



DIRECTIONS: In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option:

(A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)

(B) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)

(C) Assertion (A) is true but reason (R) is false.

(D) Assertion (A) is false but reason (R) is true.

REASON ASSERTION BASED **(1*2=2)**

19. **Assertion:** The probability of getting a bad egg in a lot of 400 is 0.035. The number of bad eggs in the lot is 14.

Reason: If the probability of an event is p, the probability of its complementary event will be 1-p

20. **Assertion (A):** HCF of any two consecutive even natural numbers is always 2.

Reason (R): Even natural numbers are divisible by 2.

SECTION B

Directions: Questions no. Q21 to Q25 are Short answer type questions of 2 marks each.

SHORT ANSWER TYPE QUESTIONS **(2*5=10)**

21. Check whether 6^n can end with the digit 0 for any natural number n.

22. If $\tan(A+B)=\sqrt{3}$ and $\tan(A-B)=\frac{1}{\sqrt{3}}$; $0^\circ < A+B < 90^\circ$; $A > B$, find A and B.

OR

If $\sin\theta+\cos\theta=\sqrt{3}$, then prove that $\tan\theta+\cot\theta=1$

23. Find a relation between x and y such that the point (x , y) is equidistant from the points (7, 1) and (3, 5).

24. A group consists of 12 persons, of which 3 are extremely patient, other 6 are extremely honest and rest are extremely kind. A person from the group is selected at random. Assuming that each person is equally likely to be selected, find the probability of selecting a person who is
(i) extremely patient 1/4
(ii) extremely kind or honest. 3/4

OR

How many positive three-digit integers have the hundredths digit 8 and unit's digit 5? Find the probability of selecting one such number out of all three-digit numbers.

25. If α and β are zeroes of a polynomial $6x^2 - 5x + 1$ then form a quadratic polynomial whose zeroes are α^2 and β^2.

SECTION C

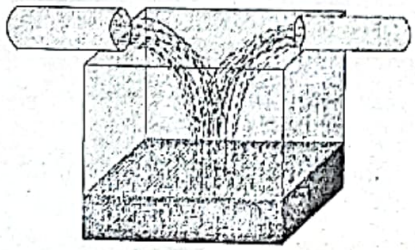
Directions: Questions no. Q26 to Q 31 are Short answer type questions of 3 marks each.

26. If the zeroes of the polynomial $x^2 + px + q$ are double in value to the zeroes of the polynomial $2x^2 - 5x - 3$, then find the values of p and q.

OR

A part of monthly hostel charges is fixed and the remaining depends on the number of days one has taken food in the mess. When a student A takes food for 20 days she has to pay Rs 1000 as hostel charges whereas a student B, who takes food for 26 days, pays Rs 1180 as hostel charges. Find the fixed charges and the cost of food per day.

27. Two water taps together can fill a tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.



28. Prove that $3 + 2\sqrt{5}$ is irrational.

29. Prove the given identity, where the angles involved are acute angles for which the expressions are defined.

$$\frac{\cos A + \sin A - 1}{\cos A - \sin A + 1} = \text{cosec } A + \cot A$$

OR

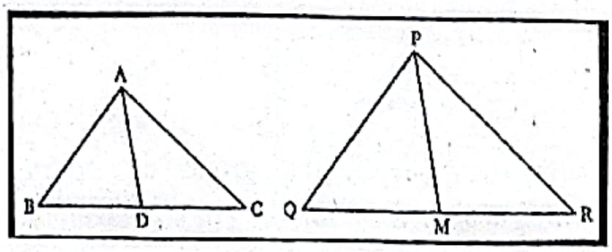
Prove the given identity, where the angles involved are acute angles for which the expressions are defined.

$$\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sec A + \tan A$$

30. On a particular day, Vidhi and Unnati couldn't decide on who would get to drive the car. They had one coin each and flipped their coin exactly three times.
The following was agreed upon:
1. If Vidhi gets two heads in a row, she would drive the car HH
2. If Unnati gets a head immediately followed by a tail, she would drive the car. HT

HHT

31. Sides AB and BC and median AD of a triangle ABC are respectively proportional to sides PQ and QR and median PM of ΔPQR (see in given figure). Show that ΔABC ~ ΔPQR.



SECTION D

Directions: Question 32-35 are long answer type questions of 5 marks each

32. Draw the graph of the following equations:

$$2x - y = 1, x + 2y = 4$$

Find the solution of the equations from the graph and shade the triangular region formed by the lines and the y-axis.

OR

The paths traced by two air planes were graphically drawn to analyze the time period of meeting point. If these paths are labeled on the Cartesian plane with coordinates, then these paths are represented in following manner:

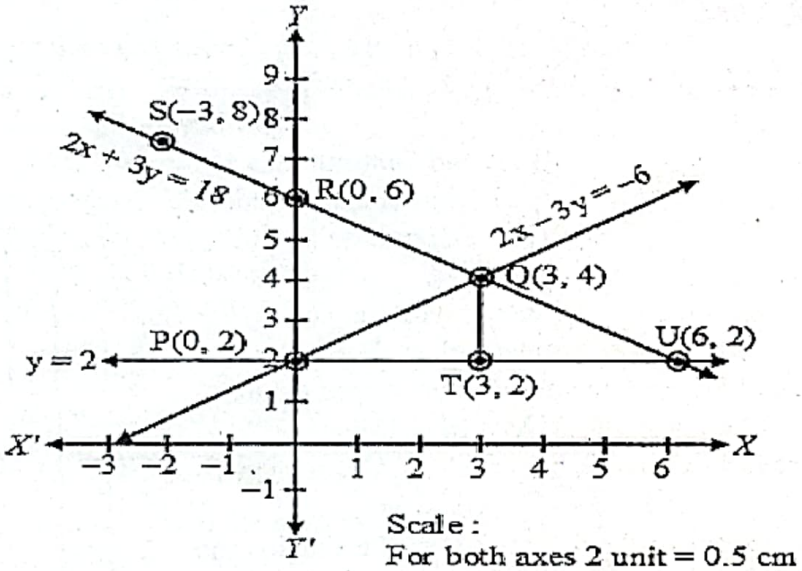
Now evaluate and find following

a) Condition for intersecting lines with standard form

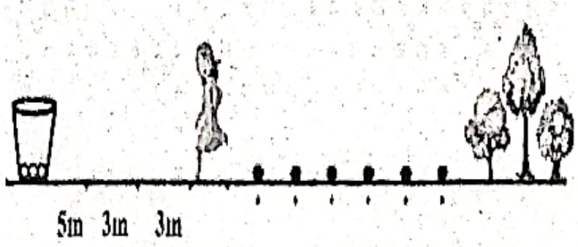
b) The area bounded by $x = 0$, $2x + 3y = 18$, and $y = 2$

c) Solution table for $y = 0$, $2x - 3y = -6$ with one two solutions

d) Perpendicular distance of the point Q from X- axis.

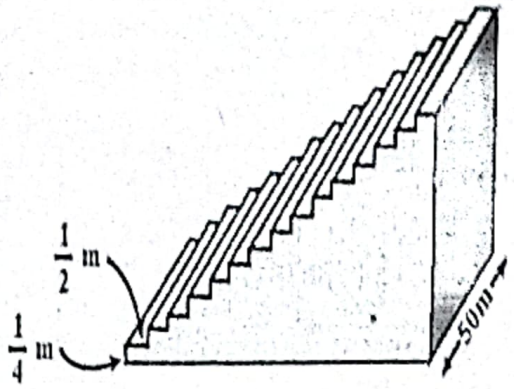


33. In a potato race, a bucket is placed at the starting point, which is 5 m from the first potato, and the other potatoes are placed 3 m apart in a straight line. There are ten potatoes in the line. A competitor starts from the bucket, picks up the nearest potato, runs back with it, drops it in the bucket, runs back to pick up the next potato, runs to the bucket to drop it in, and she continues in the same way until all the potatoes are in the bucket. What is the total distance the competitor has to run?



OR

A small terrace at a football ground comprises of 15 steps each of which is 50 m long and built of solid concrete. Each step has a rise of 1 /4 m and a tread of 1 /2 m. Calculate the total volume of concrete required to build the terrace



34. The converse of the Basic Proportionality Theorem (BPT) is:
If a line divides two sides of a triangle in the same ratio, then the line is parallel to the third side.

Prove the converse of given statement.

35. **Solve for x:**

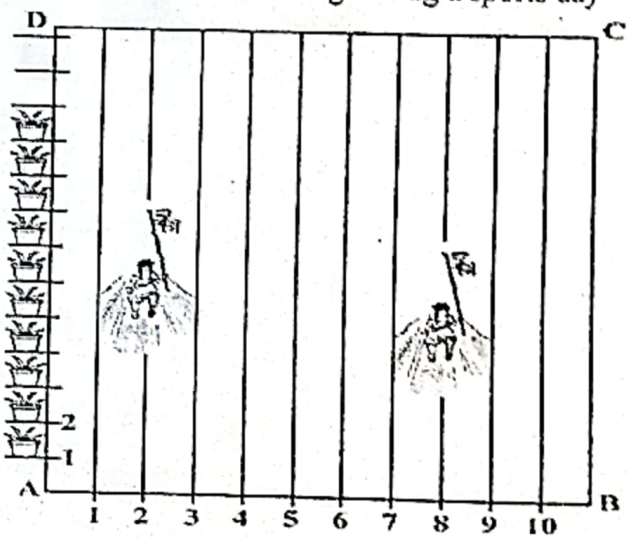
$$\frac{x-1}{2x+1}+\frac{2x+1}{x-1}=\frac{5}{2}$$

with the conditions $x \neq -\frac{1}{2}, 1$.

CASE STUDY

(4*3=12)

36. **SDG 11: Sustainable Cities and Communities.:**
To enhance our community's engagement in sustainable practices, our school is organizing a sports day with activities designed to promote well-being and environmental responsibility.
In order to conduct sports day activities in your school, lines have been drawn with chalk powder at a distance of 1 m each in a rectangular shaped ground ABCD. 100 flower pots have been placed at the distance of 1 m from each other along AD, as shown in the following figure.
Niharika runs (1/4)th distance AD on the 2nd line and posts a green Flag. Preet runs (1/5) th distance AD on the eighth line and posts are red flags. Taking A as the origin AB along x-axis and AD along y-axis, answer the following questions:



(i) Find the coordinates of the green flag.
(ii) Find the distance between the two flags.
(iii) If Rashmi has to post a blue flag exactly halfway between the line segment joining the two flags, where should she post her flag?

Or

If Joy has to post a flag at one fourth distance from the green flag, in the line segment joining the green and red flags, then where should he post his flag?

37. Case study II -The Gift of Life – Understanding Blood Groups

During a **health awareness campaign** organized by students in Sweta's school, the importance of **blood donation** was highlighted. The school invited doctors to explain how different **blood groups** play a vital role in saving lives during emergencies like surgeries, accidents, or childbirth.

The doctor explained: "Blood groups are inherited and are classified based on the antigens present on the surface of red blood cells. For a successful blood transfusion, it's important to match blood types properly. Mismatched transfusions can be life-threatening. As part of the awareness program, students conducted a survey of 50 people to understand the distribution of blood groups in their community.

Here are the results:

- **21 people had Type O blood**
- **5 people had Type B blood**
- **22 people had Type A blood**
- **2 people had Type AB blood**

Answer the following Questions:

- (i) What is the probability that a randomly selected person has **Type O** blood?
- (ii) What is the probability that a person has **Type A or Type B** blood?
- (iii) What is the probability that a person has **neither Type A nor Type O** blood?

OR

What is the probability that a person **does not have Type AB** blood?

38. Case-Based Question – Auditorium Seating for DPSJMUN

Delhi Public School (DPS) is hosting its prestigious **Model United Nations Conference (DPSJNUN)** in the school auditorium. As part of the preparations, the school management decides to renovate the **auditorium seating** to accommodate the large number of student delegates, faculty, and guests attending the international event.

To give the venue a modern look, the chairs are arranged in **concentric circular rows**, such that:

- **The first circular row has 30 seats.**
- **Each next row increases by 10 seats** compared to the previous one.

The auditorium must seat at least **1500 people** for the event.

Answer the following:

I) If the first row has 30 seats and each subsequent row increases by 10 seats, **how many seats will be there in the 10th row?**

II) **If 17 rows are arranged, how many seats will be there in the middle row?**

III) If the total number of seats required is **1500, how many rows** should be constructed to accommodate all the guests?

OR

How many seats are **still left to be placed after the 10th row?**